

RED EARTH, AB, Canada

WMO: 712460

Lat: 56.5519N

Lon: 115.2778W

Elev: 1791

StdP: 13.77

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.3	-25.8	-39.4	0.6	-32.1	-32.4	0.9	-25.7	16.6	30.4	14.2	25.5	2.2	150	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	82.5	62.4	79.1	60.5	75.9	59.0	65.1	76.7	63.0	74.3	61.2	72.0	7.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	85.5	69.2	58.9	79.3	66.7	57.1	74.2	64.8	31.0	76.6	29.4	74.6	28.1	72.1	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	13.5	11.8	DB	-37.7	88.1	6.3	2.8	-42.3	90.1	-46.0	91.7	-49.5	93.3	-54.1	95.4	(4)
(5)			WB	-38.1	68.0	6.2	3.4	-42.6	70.4	-46.2	72.4	-49.7	74.3	-54.2	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.8	6.1	11.5	21.7	37.4	49.2	57.2	61.3	58.5	49.2	37.3	18.7	8.0		
	DBStd	22.83	17.09	14.37	15.02	9.82	8.49	6.03	5.22	6.08	7.75	8.20	12.99	15.92		
	HDD50	6581	1361	1077	876	387	117	11	8	104	400	938	1301			
	HDD65	11063	1826	1497	1341	830	491	243	137	213	474	857	1388	1766		
	CDD50	1040	0	0	0	8	92	227	352	273	81	7	0	0		
	CDD65	46	0	0	0	0	1	8	23	13	1	0	0	0		
	CDH74	1105	0	0	0	5	88	230	438	281	61	2	0	0		
	CDH80	221	0	0	0	0	18	37	93	62	11	0	0	0		
	WSAvg	5.2	4.5	4.6	5.6	6.3	6.0	5.2	4.9	4.7	5.1	5.6	4.8	4.6		
	PrecAvg	19.1	1.1	0.7	0.8	0.9	1.8	3.5	3.3	2.2	1.6	1.1	1.1	0.9		
(15) Precipitation	PrecMax	25.6	2.3	1.2	1.7	1.8	3.5	6.2	5.8	4.2	4.2	2.6	2.2	2.3		
	PrecMin	12.5	0.3	0.2	0.2	0.2	0.6	1.2	1.9	0.7	0.2	0.5	0.4	0.3		
	PrecStd	3.1	0.5	0.3	0.3	0.4	0.7	1.3	1.1	0.8	0.7	0.5	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.6	47.3	56.0	71.7	82.3	84.4	86.7	86.2	81.4	69.5	51.3	44.6		
		MCWB	38.3	38.0	41.6	50.2	56.8	61.0	65.3	64.7	61.3	53.2	42.5	36.9		
	2%	DB	40.3	42.2	50.0	63.8	75.6	79.8	82.5	81.0	74.3	60.2	44.7	39.4		
		MCWB	35.6	35.1	39.0	46.1	53.9	58.6	63.7	63.0	58.2	47.7	37.9	34.6		
	5%	DB	36.0	37.7	46.4	59.1	71.6	76.3	78.9	76.6	69.0	55.2	40.2	35.0		
		MCWB	32.6	32.1	37.1	43.7	52.3	57.3	61.8	61.3	55.3	44.6	35.2	31.3		
	10%	DB	31.1	33.3	42.6	54.7	67.4	72.6	75.4	72.9	64.2	51.3	36.1	30.7		
		MCWB	28.5	29.3	34.7	41.4	50.1	56.2	60.3	59.6	52.9	42.3	32.7	28.4		
	0.4%	WB	38.8	38.8	42.9	50.9	59.9	64.2	68.9	68.7	62.7	54.1	43.1	37.7		
		MCDB	44.5	45.3	53.7	69.3	74.5	76.8	81.5	80.5	77.8	67.2	50.7	43.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.9	35.4	40.0	47.0	56.1	61.5	65.9	64.8	59.4	48.8	38.7	34.8		
		MCDB	39.9	41.0	48.7	61.3	72.1	74.6	77.7	76.2	70.7	58.6	43.9	39.3		
	5%	WB	32.6	32.4	37.5	44.5	53.5	59.5	63.6	62.7	56.7	45.6	35.6	31.6		
		MCDB	35.9	37.7	45.5	57.6	68.3	71.3	74.7	73.6	67.1	53.2	39.6	34.8		
	10%	WB	28.4	29.2	35.0	42.1	51.3	57.8	61.8	60.7	53.9	43.2	32.9	28.2		
		MCDB	31.0	33.3	42.1	53.4	64.6	69.3	72.1	70.6	62.0	50.2	35.9	30.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.7	24.5	24.1	23.1	25.4	23.7	23.1	23.4	22.7	18.3	16.6	18.3		
		MCDBR	21.5	28.7	25.8	31.2	32.6	31.6	30.5	30.4	31.7	26.4	19.9	19.5		
		MCWBR	17.1	22.2	16.5	17.2	15.2	13.9	13.9	14.8	16.7	15.6	14.4	15.8		
	5% WB	MCDBR	21.0	27.4	25.0	29.2	29.6	26.3	26.7	27.4	29.7	24.7	18.5	19.6		
		MCWBR	16.9	21.4	16.4	16.6	14.3	12.6	13.5	14.5	16.4	15.3	13.8	16.3		
(40) Clear-Sky Solar Irradiance	taub		0.226	0.244	0.271	0.326	0.342	0.352	0.358	0.348	0.311	0.283	0.241	0.218		
	taud		2.312	2.356	2.391	2.394	2.431	2.431	2.432	2.436	2.558	2.592	2.443	2.238		
	Ebn at Noon		227	270	288	284	285	282	278	273	272	254	223	196		
	Edh at Noon		15	22	27	32	33	34	33	30	23	17	13	12		
(44) All-Sky Solar Radiation	RadAvg		217	530	1017	1493	1776	1818	1794	1465	1006	538	232	141		
	RadStd		19	45	79	127	156	146	108	110	83	75	22	11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	+4.10	+5.80	N/A	N/A	N/A	N/A	N/A	+166	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+101	N/A

Nomenclature: See separate page